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Fastener for garment

Abstract

A fastener for use with undergarment, including a first tape having a body provided with a plurality of first connectors; a second tape having a body provided with a plurality of second connectors for connecting with the first connectors in the first tape; and each of the first and second tapes includes an attachment portion having a retainer adapted for connecting with the undergarment, the first and second connectors are elastic; and a method of forming the same.

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Background/Summary

(1) The present invention relates to a fastener for use with clothing to close off openings, for example particularly, but not exclusively, a hook and eye fastener for undergarment.

BACKGROUND OF THE INVENTION

(2) Conventional hook and eye fastener is a type of closure commonly used in clothing and sewing projects. It consists of two separate pieces: a small hook and a small eye. The hook is typically sewn onto one side of a garment or piece of fabric while the eye is sewn onto another side. The hook is then inserted into the eye to create a secure closure. This type of fastener is often used on the back of dresses, skirts, and blouses. It is a simple and effective alternative to close a garment other than buttons or zippers.

(3) With the eye connector lying flat on the upper surface of an eye tape, it can be difficult for user to insert the hook connector sewed to the hook tape, especially when it is done without looking. It is relatively common to use at least two pairs of hook and eye fasteners to securely close off an opening. It can be difficult to fastener one pair of hook and eye fastener blind folded, not to mention two.

(4) The problem is more profound with the less able such as an elderly who has weak grip and shaky hands. The challenges in using hook and eye fasteners are shared by the disabled and young children.

(5) Hook and eye connectors are usually made from wire or plastic. These materials are relatively hard and may cause discomfort to the skin of a user. This is particularly the case when they are used with undergarments. Cushions have been applied to mask the hardness of the hook and eye

connectors from the skin of a user but the fastener becomes very bulky. The bulkiness affects the aesthetic appearance of the undergarment and makes it difficult to conceal.

(6) The invention seeks to eliminate or at least to mitigate such shortcomings by providing a fastener for use with garments.

SUMMARY OF THE INVENTION

(7) In a first aspect of the invention there is provided a fastener for use with undergarment, comprising a first tape having a body provided with a plurality of first connectors; a second tape having a body provided with a plurality of second connectors for connecting with the first connectors in the first tape; and each of the first and second tapes includes an attachment portion having a retainer adapted for connecting with said undergarment; wherein the first and second connectors are self restorable; preferably, the attachment portion is provided at a first end of the tape; more preferably, the retainer of the attachment portion comprises a space defined by upper and lower layers of the attachment portion, the space is configured to receive a part of said undergarment; yet more preferably, the first connectors are provided adjacent the attachment portion. It is preferable that the first connectors and the attachment portion are in connection with a base layer; advantageously, the base layer is formed from a layer of stretchable material; more advantageously, the attachment portion includes a lamination of two layers of materials with different stretchabilities. Yet more advantageously, the first tape includes a gripper portion provided opposite to the attachment portion for manipulating by fingers of a user. In a preferred embodiment, the gripping portion is devoid of the first connectors; preferably, at least a part of the gripping portion includes a layer of stretchable material; preferably, the base layer is movable from a first position in which the first connectors are concealed, to a second position in which the first connectors are exposed; more preferably, the base layer is fixed in position by way of adhesive; it is preferable that the base layer is fixed to the first connection via a support layer; preferably, the first connectors are supported on a backing sheet; more preferably, the attachment portion is sewed onto a backing sheet and the base layer. It is preferable that the first connector comprises self restorable hook connector and the second connector comprises self restorable loop connector for engaging the hook connector. Preferably, the first and second tapes are Velcro™ tapes.

(8) In a second aspect of the invention there is provided a method of manufacturing a first tape as claimed in claim 1 comprising the steps of: providing a layer of first self restorable connectors; attaching the layer of first self restorable connectors onto a support layer; attaching a base layer to at least one of the layer of first self restorable connectors and the support layer at three sides of the base layer; moving the base layer from a first position to a second position to simultaneously expose the layer of first self restorable connectors, place the base layer under the support layer and conceal the three sides at which the base layer is attached to the layer of first self restorable connectors and the support layer; preferably, the method further comprises the step of forming a gripping portion adjacent the layer of first self restorable connectors.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The invention will now be more particularly described, by way of example only, with reference to the accompanying drawings, in which:

(2) FIG. 1 is an illustrative perspective view of the formation of a hook tape of a first embodiment of a fastener in accordance with the invention;

(3) FIG. 2A is an illustrative cross-sectional view of the hook tape in FIG. 1 at a first state;

(4) FIG. 2B is an illustrative cross-sectional view of the hook tape in FIG. 1 at a second state;

(5) FIG. 3A is a perspective view of the hook tape in FIG. 2B;

- (6) FIG. 3B is a side view of the hook tape in FIG. 2B;
- (7) FIG. 4 is an illustrative perspective view of the formation of a loop tape of the first embodiment of the fastener in accordance with the invention;
- (8) FIG. 5A is an illustrative cross-sectional view of the loop tape in FIG. 4 at the first state;
- (9) FIG. 5B is an illustrative cross-sectional view of the loop tape in FIG. 4 at its second state;
- (10) FIGS. 6A and 6B are illustrative perspective views of the loop tape in FIG. 4;
- (11) FIG. 6C is an illustrative side view of the loop tape in FIG. 4;
- (12) FIG. 7A is an illustrative perspective view showing the hook and loop tapes of the first embodiment of the fastener being fastened;
- (13) FIG. 7B is an illustrative perspective view showing the fastener in FIG. 7A with a gripping portion being raised;
- (14) FIG. 8 is an illustrative perspective view of the formation of a hook tape of the second embodiment of the fastener in accordance with the invention;
- (15) FIG. 9A is an illustrative cross-sectional view of the hook tape in FIG. 8;
- (16) FIG. 9B is an illustrative perspective view of the hook tape in FIG. 8;
- (17) FIG. 9C is an illustrative side view of the hook tape in FIG. 8;
- (18) FIG. 10 is an illustrative perspective view of the formation of a loop tape of the second embodiment of the fastener in accordance with the invention;
- (19) FIG. 11A is an illustrative cross-sectional view of the loop tape in FIG. 10;
- (20) FIG. 11B is an illustrative perspective view of the loop tape in FIG. 10;
- (21) FIG. 11C is an illustrative side view of the loop tape in FIG. 10;
- (22) FIG. 12A is an illustrative perspective view showing the hook and loop tapes of the second embodiment of the fastener being fastened; and
- (23) FIG. 12B is an illustrative perspective view showing the fastener in FIG. 12A with a gripping portion raised;

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

- (24) Referring to FIGS. 1 to 12B, there is shown two embodiments of a fastener **100** and **200** in accordance with the invention. The fastener **100** or **200** may be used with and undergarment to close off openings. More specifically, it is used with a brasserie.
- (25) The fastener **100** or **200** includes a first, hook tape **101/201**, and a second, loop tape **102/202**, that are reversibly interconnectable. When used with a brasserie, the hook tape **101/201** is attached, for example by way of a seam, to one arm of the brasserie and the loop tape **102/202** is attached, for example by way of a seam, to another arm of the brasserie. Another possible way of attaching the hook and loop tapes **101/102/201/202** to respective arms of the brasserie may involve adhesive or ultrasonic welding. When fastened, hook and loop connectors **101A/201A/102A/202A** on the hook and loop tapes **101/102/201/202** connect the arms of the brasserie to form a complete loop.
- (26) The hook tape **101/201** of both of the first and second embodiments includes a plurality of hook connectors **101A/201A** extending from a backing sheet **101B/201B**, which collectively forms a layer of hook connectors. The backing sheet **101B/201B** are supported on a base layer **101C/201C**. More specifically the layer of hook connectors is attached, directly or indirectly and possibly by way of adhesive, to the base layer **101C/201C** for support. An attachment portion **101D/201D** is provided at one end of the hook tape **101/201** and adjacent the hook connectors **101A/201A**. A gripping portion **101E/201E** is provided at another end of the hook tape **101/201** and located opposite to the attachment portion **101D/201D**. The gripping portion **101E/201E** and the attachment portion **101D/201D** are devoid of any hook connectors **101A/201A**. A window is defined between the gripping portion **101E/201E** and the attachment portion **101D/201D** through which the hook connectors **101A/201A** are exposed. The attachment portion **101D/201D** includes a retainer **101D1/201D1** which is an opened end dimensioned to receive the arm of a brasserie.
- (27) The loop tape **102/202** of both of the first and second embodiments include a plurality of loop connectors **102A/202A** extending from a backing sheet **102B/202B**, which collectively forms a

layer of loop connectors. The layer of loop connectors are supported on a base layer **102C/202C**. More specifically the layer of loop connectors is attached, directly or indirectly and possibly by way of adhesive, to the base layer **102C/202C** for support. An attachment portion **102D/202D** is provided at one end of the loop tape **102/202** and adjacent the loop connectors **102A/202A**. The attachment portion **102D/202D** is devoid of any loop connectors **102A/202A**.

(28) The hook connectors **101A/201A** and the loop connectors **102A/202A** are made of pliable, elastic and/or self-restorable material. An example would be a Velcro™ hook and loop connectors.

(29) With reference to FIGS. 7A and 7B as well as FIGS. 12A and 12B, the hook tape **101/201** is placed on top of the loop tape **102/202** and the two are pressed against one another so as to allow the hook connectors **101A/201A** on the hook tape **101/201** to engage, by way of hooking, onto the loop connectors **102A/202A** of the loop tape **102/202**. Overlapping between the hook tape **101/201** and the loop tape **102/202** is adjustable lengthwise so as to adjust the overall length of the fastener **100/200**, hence the overall length of the connected arms of the brasserie. The greater the degree of the lengthwise overlapping between the hook and loop tapes **101/201/102/202**, the shorter the overall length of the fastener **100/200**.

(30) FIGS. 1 to 3B show a hook tape **101** of the first embodiment of the fastener.

(31) Referring to FIG. 1, there is shown an elongated tape that is cut into hook tapes **101** of preferred dimension. As can be seen from the FIG. 1, the hook connectors **101A** are concealed by a base layer **101C** which has three sides attached to corresponding three sides of the support layer **101G/backing sheet 101B** to form a pocket with an opened end. Through the opened end, the base layer **101C** is moved or flipped to simultaneously reveal the hook connectors **101A** and be placed underneath the support layer **101G**.

(32) In more detail, as shown in FIGS. 2A to 3B, the backing sheet **101B** is placed over a support layer **101G** and occupies the middle portion of the support layer **101G**. One end of the hook tape **101** is provided with a gripping portion **101E**. The gripping portion **101E** is made of stretchable or elastic material and it is devoid of hook connectors **101A**. The gripping portion **101E** is fixed in position by attaching to an upper surface of the backing sheet **101B** at one end portion and an upper surface of the support layer **101G** at another end portion. Such attachment may be by way of a seam or adhesive. In this specific embodiment, one end of the gripping portion **101E** is sewed onto the backing sheet **101B** and the support layer **101G** while the other end of the gripping portion **101E** is ultrasonically welded to the upper surface of the support layer **101G** which is not covered by the backing sheet **101B**. The attachment portion **101D** is provided at another end of the hook tape **101** opposite the gripping portion **101E**, thereby defines a window through which the hook connectors **101A** are exposed. The attachment portion **101D** includes a retainer **101D1** defined by a gap between a folded layer of fabric **101D11** and a corresponding end portion **101D12** of the support layer **101G** which is not covered by the backing sheet **101B**. The fabric **101D11** is a lamination of two layers of fabric with different stretchabilities. A first layer being elastic and stretchable and a second layer being rigid and inelastic to offer support. This fabric **101D11** along with the end portion **101D12** of the support layer **101G** forms a retainer **101D1** in the form of an opened end dimensioned to receive an arm of a brasserie. The fabric **101D11** is sewed onto the backing sheet **101B** and the support layer **101G**.

(33) In FIG. 2A, the base layer **101C** is placed above and covers the attachment portion **101D**, the gripping portion **101E** and the hook connectors **101A**. The base layer **101C** is attached to the three sides of the layers underneath it to form a pocket. By flipping the base layer **101C** inside out, simultaneously, the base layer **101C** is placed under the support layer **101G** whilst the hook connectors **101A**, the gripping portion **101E** and the attachment portion **101D** are revealed. At the same time, the three edges where the base layer **101C** is attached to the rest of the hook tape **101** are concealed as shown in FIGS. 2B to 3B. The base layer **101C** may be attached to the support layer **101G** in a further step by way of adhesive.

(34) Referring to FIGS. 4 to 6C, there is shown a first embodiment of the loop tape **102**.

(35) In FIG. 4, there is shown an elongated tape that is cut into loop tapes **102** of preferred dimension. As can be seen from the FIGS. 4 and 5A, the loop connectors **102A** are concealed by a base layer **102C** with three sides attached to corresponding three sides of the rest of the fastener **100** to form a pocket with an opened end. Through the opened end, the base layer **102C** is moved or flipped to reveal the loop connectors **102A** and being placed underneath the support layer **102G** simultaneously.

(36) In more detail, as shown in FIGS. 4 to 6C, the loop connectors **102A** and its backing sheet **102B** is placed over a support layer **102G** and occupies most of an upper surface of the support layer **102G** except an end portion where the attachment portion **102D** is provided. The attachment portion **102D** includes a retainer **102D1** defined by a folded layer of fabric **102D11** and a corresponding end portion **102D12** of the support layer **102G**. The fabric **102D11** is a lamination of two layers of fabric with different stretchabilities. An upper layer being elastic and stretchable and a lower layer being rigid and inelastic to offer support. This fabric **102D11** along with the end portion **102D12** of the support layer **102G** forms the retainer **102D** in the form of an opened end dimensioned to receive an arm of a brasserie. One end of the fabric **102D11** is sewed onto the backing sheet **102B** and the support layer **102G**.

(37) In FIG. 5A to 6C, the base layer **102C** is placed above and covers the attachment portion **102D** and the loop connectors **102A**. The base layer **102C** is attached to the three sides of the layers underneath it, which includes the backing sheet **102B** and the support layer **102G** to form a pocket. By flipping the base layer **102C** inside out, the base layer **102C** is placed under the support layer **102G**, the loop connectors **102A** and the attachment portion **102D** are revealed simultaneously while three edges at which the base layer **102C** is attached to the rest of the loop tape **102** are concealed by the base layer **102C**. The base layer **102C** may be attached to the support layer **102G** in a further step by way of adhesive as shown.

(38) With reference to FIGS. 7A and 7B, the gripping portion **101E** offers a non-fastening portion that is devoid of any hook connectors **101A**. The gripping portion **101E** permits easy disconnection of the hook tape **101** from the loop tape **102** by offering a part to be held by fingers of a user for pulling the hook tape **101** from the hook tape **101** after the two are fastened. It also makes it easy to bring the hook tape **101** and the loop tape **102** together. In another embodiment, the gripping portion may be provided with the loop tape **102**.

(39) FIGS. 8 to 12B show a second embodiment of the fastener in accordance with the invention. The construction of the hook and loop tapes **201/202** in the second embodiment of the fastener is slightly different from that in the first embodiment as an adaptation for suiting different needs.

(40) Referring to FIG. 8, there is shown an elongated tape that is cut into hook tapes **201** of preferred dimension. As can be seen from the FIG. 8, the hook tape **201** is a lamination of various layers of material including a layer of hook connectors **201A** supported by a backing sheet **201B** which is positioned above a base layer **201C**. The backing sheet **201B** and the base layer **201C** are attached to one another via a layer of heat sensitive adhesive material **201H**.

(41) In more detail, as shown in FIGS. 9A to 9C, the hook connectors **201A** occupy a middle portion of backing sheet **201B** and the overall hook tape **201**. The backing sheet **201B** is of the same dimension as the base layer **201C** for supporting a gripping portion **201E** at one end and an attachment portion **201D** at another end opposite that of the first mentioned end. The gripping portion **201E** is a lamination of two layers of fabric with different stretchabilities. A first layer of the fabric is a stretchable or elastic material supported by a second layer that is relatively more rigid and inelastic. The gripping portion **201E** is devoid of hook connectors **201A** and is fixed in position by adhesively attached to the backing sheet **201B**. The attachment portion **201D** and the gripping portion **201E** define a window through which the hook connectors **201A** are exposed. The attachment portion **201D** includes a retainer **201D1** defined by a folded layer of fabric. The fabric **201D11** is a lamination of two layers of fabric with different stretchabilities. First layer is elastic and stretchable and a second layer being rigid and inelastic to offer support. The folded layer forms

an opened end dimensioned to receive an arm of a brasserie. Lower portion of the folded layer is attached to the backing layer **201B** by adhesive while the upper layer forms an upper surface of the fastener **200** along with the hook connectors **201A** and the gripping portion **201E**.

(42) Referring to FIGS. **10** to **11C**, there is shown a loop tape **202** of a second embodiment of the fastener **200**. As can be seen from the figures, the loop tape **202** is a lamination of various layers of material including a layer of loop connectors **202A** placed on top of a backing sheet **202B** which is positioned above a base layer **202C**. The backing sheet **202B** and the base layer **202C** are attached to one another by way of a layer of heat sensitive adhesive **202H**.

(43) In FIG. **10**, there is shown an elongated tape that is cut into loop tapes **202** of preferred dimension. The backing sheet **202B** along with the attachment portion **202D** occupied the entire upper surface of the base layer **202C**. The attachment portion **202D** includes a retainer **202D1** defined by a folded layer of fabric **202D11**. The fabric **202D11** is a lamination of two layers of fabric with different stretchabilities. First layer is elastic and stretchable and second layer is relatively rigid and inelastic to offer support. The folded layer forms an opened end dimensioned to receive an arm of a brasserie. Lower part of the folded layer is attached to the base layer **202C** by adhesive while the upper layer forms an upper surface of the fastener which is devoid of any loop connectors **202A**. The loop connectors **202A** and the upper layer of the attachment portion **202D** forms an upper surface of the fastener **202**.

(44) With reference to FIGS. **12A** and **12B**, the gripping portion **201E** of the hook tape **201** offers a gripping portion that is devoid of any hook connectors **201A**. The gripping portion **201E** permits easy disconnection of the hook tape **201** from the loop tape **202** by offering a part to be held by fingers of a user for pulling the hook tape **201** from the loop tape **202** after the two are fastened. It also makes it easy to bring the hook tape **201** and the loop tape **202** together. In another embodiment, the gripping portion may be provided with the loop tape **202**.

(45) The invention has been given by way of example only, and various other modifications of and/or alterations to the described embodiment may be made by persons skilled in the art without departing from the scope of the invention as specified in the appended claims.

Claims

1. A fastener for use with undergarment, comprising: a first tape having a body provided with a plurality of first connectors; a second tape having a body provided with a plurality of second connectors for connecting with the first connectors in the first tape; and each of the first and second tapes includes an attachment portion having a retainer adapted for connecting with said undergarment; wherein the first and second connectors are self restorable; wherein the attachment portion of each of the first and second tapes is provided at a first end of each of the first and second tapes; and the retainer of the attachment portion comprises a space defined by upper and lower layers of the attachment portion, the space is configured to receive a part of said undergarment.
2. The fastener as claimed in claim 1, wherein the plurality of first connectors is provided adjacent the attachment portion of the first tape.
3. The fastener as claimed in claim 2, wherein the plurality of first connectors and the attachment portion of the first tape are in connection with a base layer.
4. The fastener as claimed in claim 3, wherein the base layer is formed from a layer of stretchable material.
5. The fastener as claimed in claim 3, wherein the base layer is movable from a first position in which the first plurality of connectors is concealed, to a second position in which the first plurality of connectors is exposed.
6. The fastener as claimed in claim 3, wherein the base layer is attached to the plurality of first connectors by way of adhesive.
7. The fastener as claimed in claim 3, wherein the base layer is fixed to the plurality of first

connectors via a support layer.

8. The fastener as claimed in claim 2, wherein the first tape includes a gripper portion provided opposite to the attachment portion of the first tape for manipulating by fingers of a user.

9. The fastener as claimed in claim 8, wherein the gripping portion is devoid of the first connectors.

10. The fastener as claimed in claim 9, wherein at least a part of the gripping portion includes a layer of stretchable material.

11. The fastener as claimed in claim 1, wherein the attachment portion of each of the first and second tapes includes a lamination of two layers of materials with different stretchabilities.

12. The fastener as claimed in claim 1, wherein the plurality of first connectors is supported on a backing sheet.

13. The fastener as claimed in claim 12, wherein the attachment portion of the first tape is sewed onto said backing sheet and a base layer.

14. The fastener as claimed in claim 1, wherein the plurality of first connectors comprise self restorable hook connectors and the plurality of second connectors comprise self restorable loop connectors for engaging the hook connectors.

15. A method of manufacturing a first tape as claimed in claim 1 comprising the steps of: providing a layer of first self restorable connectors; attaching the layer of first self restorable connectors onto a support layer; attaching a base layer to at least one of the layer of first self restorable connectors and the support layer at three sides of the base layer; moving the base layer from a first position to a second position to simultaneously expose the layer of first self restorable connectors, place the base layer under the support layer and conceal the three sides at which the base layer is attached to the layer of first self restorable connectors and the support layer.

16. The method as claimed in claim 15 further comprising the step of forming a gripping portion adjacent the layer of first self restorable connectors.

17. The fastener as claimed in claim 1, wherein the first and second tapes are hook and loop tapes.

18. A fastener for use with undergarment, comprising: a first tape having a body provided with a plurality of first connectors; a second tape having a body provided with a plurality of second connectors for connecting with the first connectors in the first tape; and each of the first and second tapes includes an attachment portion having a retainer adapted for connecting with said undergarment; wherein the first and second connectors are self-restorable; wherein the attachment portion of each of the first and second tapes includes a lamination of two layers of materials with different stretchabilities.

19. A fastener for use with undergarment, comprising: a first tape having a body provided with a plurality of first connectors; a second tape having a body provided with a plurality of second connectors for connecting with the first connectors in the first tape; and each of the first and second tapes includes an attachment portion having a retainer adapted for connecting with said undergarment; wherein the first and second connectors are self-restorable; wherein the plurality of first connectors are provided adjacent the attachment portion; the first tape includes a gripper portion provided opposite to the attachment portion of the first tape for manipulating by fingers of a user; and the gripping portion is devoid of the plurality of first connectors.

20. A fastener for use with undergarment, comprising: a first tape having a body provided with a plurality of first connectors; a second tape having a body provided with a plurality of second connectors for connecting with the first connectors in the first tape; and each of the first and second tapes includes an attachment portion having a retainer adapted for connecting with said undergarment; wherein the first and second connectors are self-restorable; wherein the attachment portion is provided at a first end of each of the first and second tapes; wherein the plurality of first connectors are supported on a backing sheet; and the attachment portion of the first tape is sewn onto said backing sheet and a base layer.
